```
from google.colab import files
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Upload image
uploaded = files.upload()

# Read uploaded image
filename = list(uploaded.keys())[0]
image = cv2.imread(filename)
print("Image uploaded successfully:", filename)

# Get image size
height, width = image.shape[:2]
print("Width:", width)
print("Height:", height)

# Select 4 source points
source_points = np.float32([
    [50, 50],
    [width - 50, 50],
    [50, height - 50],
    [width - 50, height - 50]
])

# Set 4 destination points
destination_points = np.float32([
    [0, 0],
    [width, 0],
    [100, height],
    [width - 100, height]
])

# Calculate perspective transformation matrix
matrix = cv2.getPerspectiveTransform(
    source_points,
    destination_points
)

print("Perspective Matrix:")
print(matrix)

# Apply perspective transformation
perspective_image = cv2.warpPerspective(
    image,
    matrix,
    (width, height)
)

# Display original and transformed images
plt.figure(figsize=(10, 5))
```

```
plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(perspective_image, cv2.COLOR_BGR2RGB))
plt.title("Perspective Transformed Image")
plt.axis("off")

plt.show()
```

Choose Files Screenshot… 075823.png

Screenshot 2026-10-10 075823.png(image/png) - 93416 bytes, last modified: 10/10/2026 - 100% done

```
Saving Screenshot 2026-10-10 075823.png to Screenshot 2026-10-10 075823 (3).p
Image uploaded successfully: Screenshot 2026-10-10 075823 (3).png
Width: 299
Height: 196
Perspective Matrix:
[[-2.87900794e+01 -6.02822581e+01  4.45361687e+03]
 [ 0.00000000e+00 -1.18153226e+02  5.90766129e+03]
 [ 5.55036220e-17 -4.03225806e-01  1.00000000e+00]]
```



Getting started

The document you are reading is not a static web page, but an interactive environment called a **Colab notebook** that lets you write and execute code.

For example, here is a **code cell** with a short Python script that computes a value, stores it in a variable, and prints the result:

```
seconds_in_a_day = 24 * 60 * 60
seconds_in_a_day
```

```
86400
```